COER UNIVERSITY

ASSIGNMENT-3

Course: MCA | Subject: Machine Learning (MCA201T)

Submitted by: Mahi dhiman ***Roll No:*** 253028050

Q1. Define Supervised Learning. Explain the key difference between supervised learning and unsupervised learning with one example of each.

Definition

Supervised Learning is a type of machine learning in which a model is trained on a *labelled dataset*, i.e. every training example has input features (X) and a known correct output or label (y). The algorithm learns a mapping function f: X → y by minimising the error between its predictions and the true labels. Once trained, the model is used to predict the output for new, unseen inputs. It is called "supervised" because the labels act like a teacher that supervises the learning process.

Difference between Supervised and Unsupervised Learning

Basis	***Supervised Learning***	***Unsupervised Learning***
Data used	Labelled data (X, y).	Unlabelled data (only X).
Goal	Learn a mapping to predict a known target.	Discover hidden structure, patterns or groups in data.
Feedback	Direct feedback: correct label is available for every example.	No labels and no direct feedback.
Main tasks	Classification and regression.	Clustering, dimensionality reduction, association rules.
Evaluation	Accuracy, precision, recall, RMSE, etc.	Silhouette score, inertia, domain judgement.
Example	Spam filtering: emails labelled spam / not spam are used to train a classifier.	Customer segmentation: k-means groups customers by buying behaviour without predefined groups.

Q2. What are the two main types of supervised learning problems? Clearly distinguish between them and give one real-world example for each type.

The two main types of supervised learning problems are ***Classification*** and ***Regression***. In classification the output is a discrete category, while in regression the output is a continuous numerical value.

Basis	***Classification***	***Regression***
Output	Discrete class label (e.g., spam / not spam).	Continuous real number (e.g., price, temperature).
Goal	Assign an input to one of the predefined classes.	Predict a quantity.
Types	Binary and multi-class classification.	Simple, multiple and polynomial regression.
Algorithms	Logistic regression, decision tree, SVM, k-NN, Naive Bayes.	Linear regression, ridge / lasso, regression trees, random forest regressor.

Basis	***Classification***	***Regression***
Metrics	Accuracy, precision, recall, F1, ROC-AUC.	MSE, RMSE, MAE, R^2.
Real-world example	***Disease diagnosis:*** predicting whether a tumour is malignant or benign.	***House price prediction:*** predicting the price of a house from its area, location and number of rooms.

Q3. Explain the concept of Linear Regression. Write the equation of a simple linear regression model and define the terms (dependent variable, independent variable, intercept, and slope). What is the role of the residual/error term?

Concept

Linear Regression is a supervised learning algorithm that models the relationship between a continuous dependent variable and one or more independent variables by fitting a straight line (or hyperplane) to the data. The best-fit line is found using the ***Ordinary Least Squares (OLS)*** method, which minimises the sum of squared differences between the actual and predicted values.

Simple Linear Regression Equation

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- ***Dependent variable (y):*** the target / response that we want to predict (e.g., salary).
- ***Independent variable (x):*** the input or predictor used to explain y (e.g., years of experience).
- ***Intercept (β_0):*** the value of y when x = 0; the point where the line cuts the y-axis.
- ***Slope (β_1):*** the change in y for a one-unit increase in x.
- ***Error term (ε):*** the difference between the actual and the predicted value.

OLS Estimates

$$\beta_1 = \Sigma\,(x_i - mean(x))\,(y_i - mean(y)) \,/\, \Sigma\,(x_i - mean(x))^2 \qquad \beta_0 = mean(y) - \beta_1 \cdot mean(x)$$

Role of the Residual / Error Term

The residual $e_i = y_i - \hat{y}_i$ measures how far each prediction is from the actual value. It captures everything the straight-line model cannot explain, such as random noise, measurement errors and omitted variables. The model is trained by minimising the Sum of Squared Residuals ($SSR = \Sigma\, e_i^2$). Residuals are also used to check the model assumptions: they should have mean zero, constant variance (homoscedasticity), be independent, and be approximately normally distributed.

Q4. How does Logistic Regression differ from Linear Regression? Write the logistic (sigmoid) function and explain why it is suitable for binary classification. What is the decision boundary in logistic regression?

Difference from Linear Regression

Basis	***Linear Regression***	***Logistic Regression***
Problem type	Regression (continuous output).	Classification (probability of a class).
Output range	$-\infty$ to $+\infty$.	0 to 1 (a probability).
Function used	Straight line $y = \beta_0 + \beta_1 x$.	Sigmoid function applied to a linear combination of inputs.
Cost function	Mean squared error.	Log-loss (binary cross-entropy).
Estimation	Least squares.	Maximum likelihood.

Sigmoid (Logistic) Function

$$\sigma(z) = 1 / (1 + e^{-z}), \quad \text{where } z = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$$

Why it is suitable for binary classification

- It squashes any real number into the range (0, 1), so the output can be read as the probability P(y = 1 | x).
- It is S-shaped: large positive z gives a probability close to 1 and large negative z gives a probability close to 0.
- It is smooth and differentiable, which makes gradient-based optimisation easy.
- The model is linear in the log-odds: ln(p / (1 − p)) = z.

Decision Boundary

We predict class 1 if P(y = 1 | x) ≥ 0.5, otherwise class 0. Since σ(z) = 0.5 exactly when z = 0, the decision boundary is the set of points where

$$\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n = 0$$

This is a straight line (for two features) or a hyperplane (for more features) that separates the two classes. The threshold 0.5 can be changed depending on the relative cost of false positives and false negatives.

Q5. Describe how a Decision Tree works for classification. Explain the concepts of Information Gain and Gini Impurity. Why can decision trees overfit the training data, and how can this problem be reduced?

Working of a Decision Tree

A decision tree is a flowchart-like model. The ***root node*** contains the whole dataset. At each node, the algorithm selects the feature (and threshold) that splits the data into the purest possible subsets and creates branches. This recursive splitting continues until a stopping condition is met (pure node, maximum depth, minimum samples). Each ***leaf node*** gives the predicted class (the majority class). To classify a new sample, we start at the root and follow the branches according to its feature values until a leaf is reached.

Entropy and Information Gain

Entropy measures the impurity or disorder of a node:

$$Entropy(S) = -\Sigma\, p_i \log_2 p_i$$

Information Gain (IG) is the reduction in entropy after splitting on attribute A (used in ID3 / C4.5). The attribute with the highest IG is chosen:

$$IG(S, A) = Entropy(S) - \Sigma\, (|S_v| / |S|) \times Entropy(S_v)$$

Gini Impurity

Gini Impurity (used in CART) is the probability of wrongly classifying a randomly chosen sample if it is labelled according to the class distribution of the node:

$$Gini(S) = 1 - \Sigma\, p_i^2$$

Gini = 0 means a perfectly pure node (for two classes the maximum is 0.5). The split with the lowest weighted Gini is selected. ***Example:*** a node with 6 “Yes” and 4 “No” has Gini = $1 - (0.6^2 + 0.4^2) = 0.48$.

Why Decision Trees Overfit

If a tree is allowed to grow fully, it keeps splitting until it memorises every training sample, including noise and outliers. The tree becomes very deep and complex with low bias but high variance, so it performs poorly on unseen data.

How to Reduce Overfitting

- ***Pre-pruning:*** limit the maximum depth, minimum samples per split / leaf, or the number of leaf nodes.
- ***Post-pruning:*** grow the full tree and then remove branches that add little predictive power (e.g., cost-complexity pruning).
- ***Cross-validation*** to choose suitable hyperparameters.
- ***Ensemble methods*** such as Random Forest, which average many trees.

Q6. What are Ensemble Methods? Differentiate between Bagging and Boosting with suitable examples of algorithms that use each technique. Why do ensemble methods generally perform better than a single model?

Ensemble Methods

Ensemble methods combine the predictions of several base models (weak learners) to build one stronger, more accurate and more robust model. The predictions are combined by majority voting (classification) or averaging (regression).

Bagging versus Boosting

Basis	***Bagging (Bootstrap Aggregating)***	***Boosting***
Training	Models are trained independently (in parallel) on random bootstrap samples.	Models are trained sequentially; each new model focuses on the errors of the previous ones.
Main aim	Reduces variance (overfitting).	Reduces bias (and also variance).
Sample weights	All samples are equally likely.	Misclassified samples get higher weights.
Combination	Simple vote / average.	Weighted vote / weighted sum.
Base learners	Strong, high-variance learners (deep trees).	Weak learners (shallow trees / stumps).
Algorithms	Bagged decision trees, Random Forest.	AdaBoost, Gradient Boosting, XGBoost, LightGBM.

Why Ensembles Perform Better than a Single Model

- Individual models make different errors; when combined, these errors tend to ***cancel out*** (wisdom of the crowd), which lowers variance.
- Boosting turns many weak learners into one strong learner, which lowers bias.
- Ensembles are more ***robust*** to noise and outliers and less likely to overfit than a single complex model.
- Combining diverse models covers a larger hypothesis space and gives better generalisation, provided the base models are reasonably accurate and diverse.

Q7. Explain the working principle of Support Vector Machines (SVM) for binary classification. What are support vectors and the margin? Briefly describe the role of the kernel trick in SVM.

Working Principle

A Support Vector Machine finds the ***optimal separating hyperplane*** $w \cdot x + b = 0$ that divides the two classes with the ***maximum margin***. Out of the many hyperplanes that can separate the data, SVM chooses the one that is farthest from the nearest points of both classes, because a larger margin gives better generalisation. With class labels $y_i \in \{-1, +1\}$, the hard-margin problem is:

$$\textit{Minimise } \tfrac{1}{2}\|w\|^2 \quad \textit{subject to} \quad y_i\,(w \cdot x_i + b) \geq 1$$

For data that is not perfectly separable, a ***soft margin*** is used: slack variables ξ_i and a penalty parameter C allow some misclassification (large C gives a narrow margin with fewer errors; small C gives a wider margin with more tolerance).

Margin and Support Vectors

- ***Margin:*** the distance between the two parallel hyperplanes $w \cdot x + b = +1$ and $w \cdot x + b = -1$. It equals $2 / \|w\|$, so maximising the margin is the same as minimising $\|w\|$.
- ***Support vectors:*** the training points that lie exactly on the margin boundaries (closest to the hyperplane). Only these points decide the position of the hyperplane; removing any other point does not change the model.

Kernel Trick

When the data is not linearly separable in the original space, SVM implicitly maps it to a higher-dimensional space where a linear separator exists. Instead of computing the mapping $\varphi(x)$ explicitly, it uses a kernel function $K(x_i, x_j) = \varphi(x_i) \cdot \varphi(x_j)$ that directly gives the inner product in that space, which is computationally cheap. Common kernels are:

- ***Linear:*** $K(x, z) = x \cdot z$
- ***Polynomial:*** $K(x, z) = (x \cdot z + c)^d$
- ***RBF (Gaussian):*** $K(x, z) = \exp(-\gamma \|x - z\|^2)$

Q8. List and briefly explain any four-evaluation metrics used for classification problems (e.g., Accuracy, Precision, Recall, F1-Score, ROC-AUC). When is Accuracy not a good metric, and which metric would you prefer instead?

All the metrics are derived from the ***confusion matrix***, which has four values: TP (true positives), TN (true negatives), FP (false positives) and FN (false negatives).

Metric	***Formula***	***Meaning***
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Fraction of all predictions that are correct.
Precision	TP / (TP + FP)	Out of all predicted positives, how many are really positive. Important when false positives are costly (e.g., spam filter).
Recall (Sensitivity)	TP / (TP + FN)	Out of all actual positives, how many were found. Important when false negatives are costly (e.g., cancer detection).
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	Harmonic mean of precision and recall; balances both.
ROC-AUC	Area under the TPR vs FPR curve	Ability to separate the classes over all thresholds; 1 = perfect, 0.5 = random.

When is Accuracy not a good metric?

Accuracy is misleading on ***imbalanced datasets***. For example, if only 1% of transactions are fraudulent, a model that always predicts "not fraud" is 99% accurate but detects no fraud at all. Accuracy also ignores the different costs of false positives and false negatives.

Preferred Metric

Recall is preferred when missing a positive case is dangerous (disease, fraud), ***Precision*** when false alarms are costly, and the ***F1-Score*** (or PR-AUC / ROC-AUC) when a balance between both is needed on imbalanced data.

Q9. For regression problems, explain the following evaluation metrics: (a) Mean Squared Error (MSE), (b) Root Mean Squared Error (RMSE), (c) Mean Absolute Error (MAE), (d) R-squared (Coefficient of Determination). Which metric is more sensitive to outliers and why?

Let y_i be the actual value, $\hat{y}_i$ the predicted value, mean(y) the average of the actual values and n the number of samples.

Metric	***Formula***	***Explanation***
(a) MSE	$MSE = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$	Average of the squared errors. Always ≥ 0 and lower is better. Squaring penalises large errors heavily; its unit is the square of the unit of y.
(b) RMSE	$RMSE = \sqrt{MSE}$	Square root of MSE. It brings the error back to the same unit as y, so it is easier to interpret.
(c) MAE	$MAE = (1/n)\ \Sigma\ \lvert y_i - \hat{y}_i \rvert$	Average of the absolute errors. Same unit as y; all errors are treated linearly.
(d) R-squared	$R^2 = 1 - SS_{res} / SS_{tot}$ $= 1 - \Sigma\ (y_i - \hat{y}_i)^2 / \Sigma\ (y_i - \text{mean}(y))^2$	Proportion of the variance in y explained by the model. 1 = perfect fit, 0 = no better than predicting the mean; it can be negative for a very poor model.

Sensitivity to Outliers

MSE (and therefore RMSE) is the most sensitive to outliers because the errors are ***squared*** before averaging. A single large error is magnified quadratically; for example, an error of 10 adds 100 to the sum in MSE but only 10 in MAE. MAE grows only linearly with the error, so it is more robust to outliers.

Q10. Discuss any four real-world applications of Supervised Learning. For each application, mention the type of supervised learning problem (classification or regression) and a suitable algorithm that can be used.

Application	***How supervised learning is used***	***Problem type***	***Suitable algorithm***
1. Email spam detection	Emails labelled as spam / not spam are used to train a model on words, sender and links; new emails are then filtered automatically.	Classification (binary)	Naive Bayes, Logistic Regression
2. House price prediction	Past sales data (area, location, rooms, age) with known prices is used to predict the price of a new property.	Regression	Linear Regression, Random Forest Regressor
3. Medical diagnosis	Patient records and test reports with known diagnoses are used to predict diseases such as diabetes or cancer.	Classification	SVM, Decision Tree / Random Forest

Application	*How supervised learning is used*	*Problem type*	*Suitable algorithm*
4. Sales / demand forecasting	Historical sales along with season, promotions and price are used to forecast future sales or stock requirements.	Regression	Gradient Boosting (XGBoost), Linear Regression

Other applications include credit card fraud detection (classification, Random Forest), image and face recognition (classification, CNN) and temperature prediction (regression).